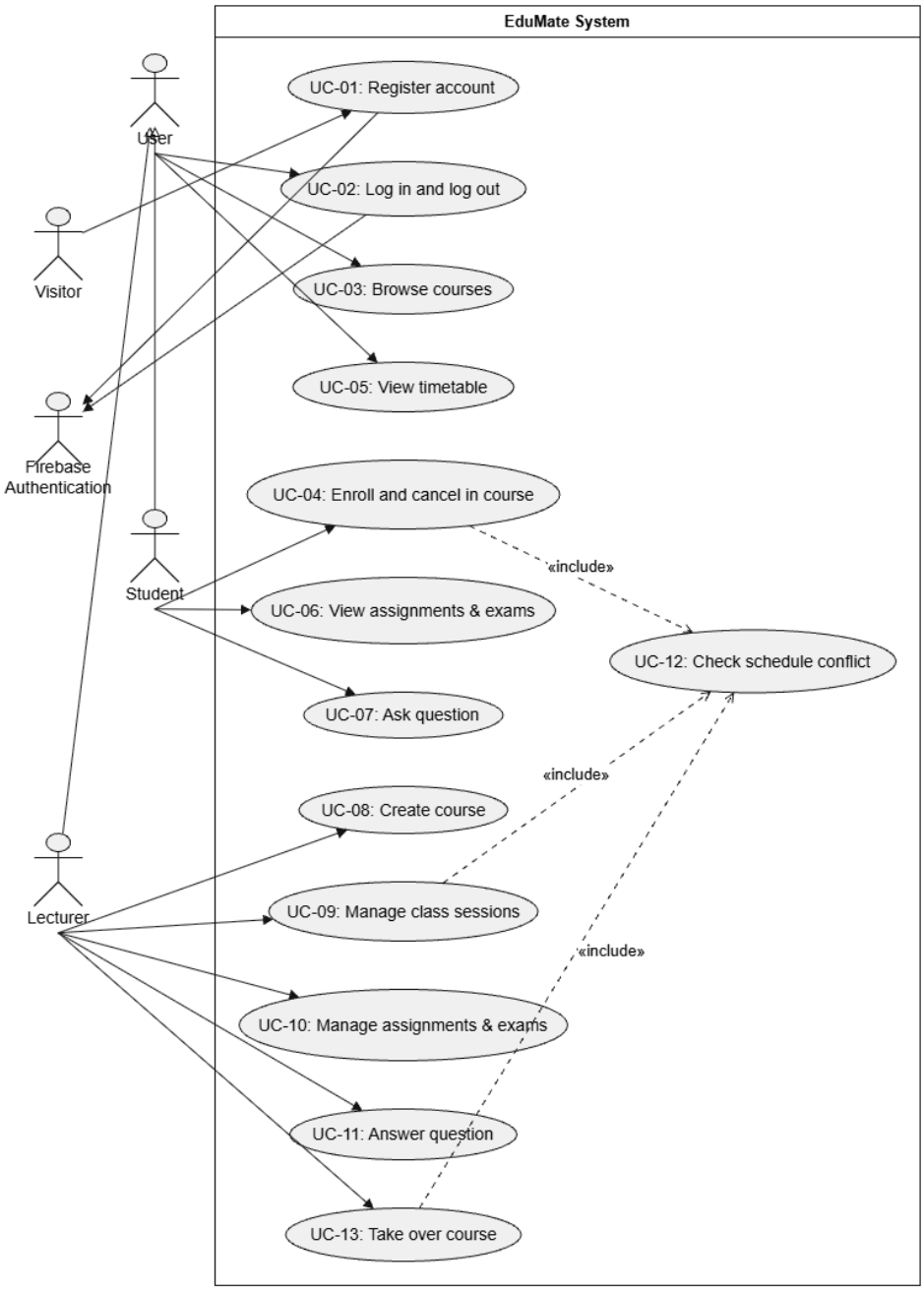
3.3 Create UML Diagrams

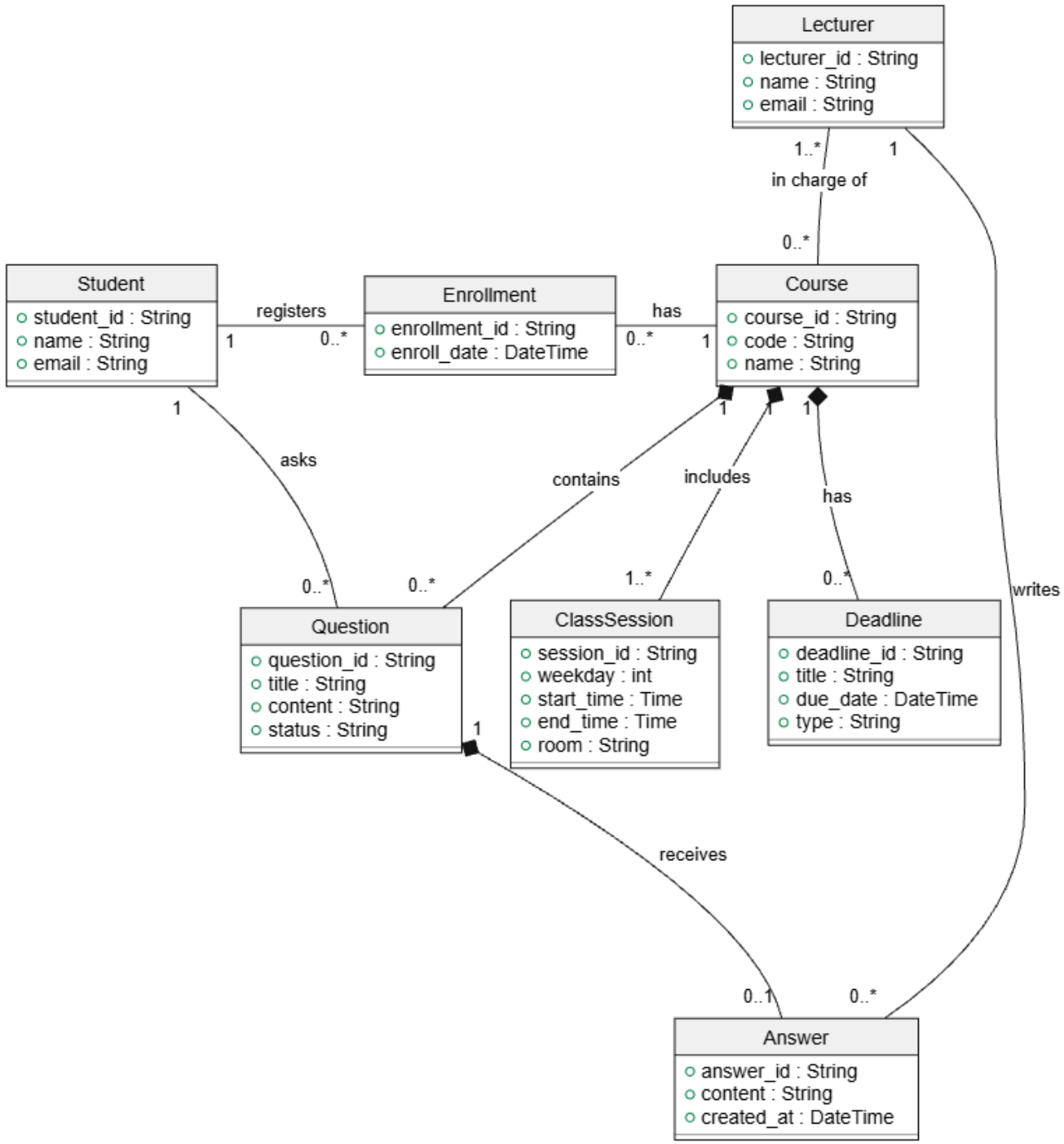
3.3.1. Use Case Diagram

The Use Case diagram provides a high-level overview of the actors within the EduMate ecosystem (Student, Lecturer) and their key interactions with the system, such as enrolling in courses, managing sessions, and asking questions.



3.3.2. Domain Class Diagram (Data & Relationships)

The Class Diagram models the core entities of EduMate and their structural relationships. It represents the backbone of the database schema, detailing attributes and relationships across Users, Courses, Class Sessions, Deadlines, and Q&A interactions.



3.3.3. Sequence Diagram: Course Enrollment & Conflict Check

To illustrate the dynamic behavior of the system, this Sequence Diagram details the execution flow of **UC-04: Enroll in a course**. It explicitly demonstrates the interaction between the React Web UI, the Python Flask API, and the Schedule System to handle HTTP requests, fetch database records, and return HTTP status codes (200 OK for success, or 409 Conflict if schedule overlapping occurs).

